



# Experimental Design for Multi-modal Take-over Request for Automated Driving

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**Abstract.** Despite recent advancement in autonomous vehicles, limitations of autonomous driving system still remain. When the autonomous driving system encounters failure, it hands over dynamic driving task to the driver. In the event of a failure of the autonomous driving system, effective take-over request should be issued according to driver's status to guarantee safety and comfort of a driver. To devise the optimal alert modalities for take-over request that can minimize the risk of failure in the autonomous driving system, the following factors were drawn out for the experimental design: 4 scenarios for take-over request through the case analysis of cause of autonomous driving disengagement; and verifiable vehicle alert modalities of visual, auditory, and tactile aspects for take-over request through the literature search and market research. Then, we have established a virtual reality simulator for experiment environment that can effectively identify driver's status based on human factors. This study will find out effective multi-modality for take-over request through an analysis of statistically significant differences according to driver's status based on human factors by drawing the quantitative/qualitative data based on scenarios for take-over request and combinations of vehicle alert modalities of visual, auditory, and tactile aspects obtained from the results of previous studies analyzed in this experiment environment.

**Keywords:** Autonomous driving system · Take-over request · Human factor Alert · Modality · Vehicle interface

## 1 Introduction

Car manufacturers in the world recently release vehicles equipped with diverse cutting-edge functions, providing safety and convenience for drivers effectively. Such cutting-edge functions are defined as the Advanced Driver Assistance System (ADAS) and cars with ADAS are called autonomous vehicles. Car makers and engineering scientists over the world focus on developing the sensor technology of autonomous vehicles. When developing an autonomous vehicle, it is important to develop high-performance sensors but it is also essential to consider HMI (Human-Machine Interaction) [1]. Amid the rising issues over autonomous vehicles these days, research on the HMI aspect has begun to be stimulated.

In autonomous vehicles, their automation levels are defined under diversified criteria; and, depending upon the levels, HMI research should be progressed differently. At the lowest level (e.g., no automation, or level 0 in [2]) of automation the human driver is in full control of the car. Cars at automation level 3 have lower human participation in driving task of driver. But, when the autonomous driving system reaches its limit, human drivers are required to involve in driving. Such an autonomous vehicle is defined as a conditional autonomous vehicle (e.g., level 3 in [2]). In full automation (e.g., level 5 in [2]), the human driver is not involved in any driving task anymore [3].

Compared to existing work [4, 8], in this present study, experimental design was implemented for the designing of a multi-modality alert capable of making request alerts in the most effective manner in the take-over situation that can happen in partial or conditional autonomous vehicles. This study seeks to design a multi-modal take-over request which can enhance the safety of autonomous vehicle by delivering take-over requests to human drivers most effectively based on take-over cases analysis, take-over scenario design, and alert modalities of visual, auditory and tactile aspects.

## **2 Take-over Case Study and Scenario Design**

### **2.1 Analysis of Cases of Autonomous Driving Disengagement**

Various companies are developing autonomous driving system, and doing real road driving tests with the license on the temporary operation of such autonomous vehicles in CA, US. The manufacturers record and present in report events of autonomous driving disengagement during their real road tests. Such reports contain the frequency of autonomous driving disengagement events, cause of disengagement, reaction time in the event of take-over request, etc. Eleven manufacturers organized the 2015 and 2016 records of test and submitted them as the autonomous driving disengagement reports to the Department of Motor Vehicles in State of California by January 1, 2017 [5–7].

In this study, 3,271 disengagement cases were analyzed in reports by 7 organizations, which seemed to have been prepared in an analyzable manner. The circumstances and causes of take-over request recorded in the reports were expressed in key words and categorized under driver, vehicle, and environment factors. The driver factor means disengagement by drivers as they felt uncomfortable due to the delay of deceleration in autonomous vehicle, failure of lane keeping control, etc. The vehicle factor means disengagement by the system based on the failure of perception, decision, control and network of the autonomous system or vehicle failure. The environment factor means autonomous driving system disengagement because of traffic participants, road, weather, traffic flow, and obstruction. As a result of statistical analysis, the driver factor accounted for 21.7% of the disengagement causes; vehicle factor, 53.4%; and environment factor, 24.8%.

### **2.2 Design of Take-over Request Scenario**

To design the optimal take-over multi-modal request alert method, take-over scenario design was implemented in this study. This study looks at only the system disengagement to explore an effective take-over request alert method.

First, the causes of take-over request can be divided into two groups at a higher level. Type A is defined as unplanned take-over circumstances and Type B, planned take-over circumstances. Type A is subdivided into autonomous system failure, vehicle failure, and unplanned ODD (Operational Design Domain) exit. Type B is classified under Planned ODD Exit, meaning that the autonomous system is ended as its planned driving task is finished. Examples of Type A and Type B take-over circumstances are shown in Table 1.

**Table 1.** Examples of take-over request according to disengagement causes.

| Type   | Cause of disengagement    | Example of take-over request                                                       |
|--------|---------------------------|------------------------------------------------------------------------------------|
| A type | Autonomous System Failure | Faulty sensor perceptions such as RADAR/LiDAR to halt autonomous driving           |
|        |                           | Localization failure to halt autonomous driving                                    |
|        | Vehicle Failure           | ECU (engine control unit) failure to lose basic vehicle functionality              |
|        | Unplanned ODD Exit        | Insufficient lane mark to halt autonomous driving                                  |
|        |                           | Construction zone not registered with route information to halt autonomous driving |
| B type | Planned ODD Exit          | Autonomous driving mode finished upon the completion of a planned route            |

For the implementation of take-over request scenario, scenario design elements were identified [9]. As in Table 2, the scenario design elements were classified based on vehicle type, driving agent, conditions of autonomous driving system and driving speed. Take-over request scenario can be established by adding the design elements to the examples of take-over request identified previously in Table 1. The examples of take-over request scenario are in Table 3.

**Table 2.** Take-over request scenario design elements.

| Classification    |                  | Design elements                                                                          |
|-------------------|------------------|------------------------------------------------------------------------------------------|
| Vehicle type      |                  | Car, truck                                                                               |
| Driving agent     |                  | Driver, system                                                                           |
| Autonomous system | Automation Level | 0, 1, 3, 4, 5 (e.g., level in [2])                                                       |
|                   | Task             | Lateral control, longitudinal control, monitor driving environment, fallback performance |
|                   | Duration         | Short time, continuous                                                                   |
| Speed range       |                  | Low, mid, high                                                                           |

**Table 3.** Examples of take-over request according to disengagement causes.

| Take-over request circumstance                                           | Scenario design element                                                  | Resulted scenario                                                                                                                                                                                  |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Perception sensor failure such as RADAR/LiDAR to halt autonomous driving | Vehicle type: Car                                                        | When the SAE L3 autonomous vehicle performs longitudinal/lateral control and monitors driving environment at 60 km/h, the system requests the driver to take over due to perception sensor failure |
|                                                                          | Driving agent: System                                                    |                                                                                                                                                                                                    |
|                                                                          | Automation level: 3                                                      |                                                                                                                                                                                                    |
|                                                                          | Task: lateral control, longitudinal control, monitor driving environment |                                                                                                                                                                                                    |
|                                                                          | Duration: Continuous                                                     |                                                                                                                                                                                                    |
|                                                                          | Speed Range: High (more than 60 km)                                      |                                                                                                                                                                                                    |

### 3 Previous Study on Alert Modality

#### 3.1 Guidance on the Alert Modalities for Non-autonomous Vehicle

Since there is no guidance on alert modalities of visual, auditory, and tactile aspects for autonomous vehicles; those on alert modalities for non-autonomous were investigated first before producing such guidance for autonomous vehicles. As a result, design elements to consider according to visual, auditory and tactile modalities were organized and such design elements can be differently applied to alert interface depending upon circumstantial urgency and a driver's status.

First of all, the visual modality-based information delivery device is one of the means to communicate information with drivers effectively. Conditions can vary according to visual display type and locating, alert color, character height, legibility, temporal characteristics, and display glare to form an alert interface. Next, the auditory modality-based information delivery device can have an alert interface by varying the conditions according to perceived urgency, perceived annoyance, loudness, distinctiveness, localization cues, and using speech. Lastly, the tactile modality information delivery device is divided into vibrotactile interface and kinesthetic interface and can have an alert interface by setting different conditions according to haptic display type, frequency band that human can detect, and period.

#### 3.2 Previous Study on Alert Modality of Autonomous Vehicle

As the roles of human drivers change in autonomous vehicles, research on new DVI (Driver-Vehicle Interaction) related to autonomous vehicles is going on actively. Although there are not enough studies on the alert methodology in take-over request circumstances, studies on take-over request time and autonomous vehicle alert interface were inspected before producing take-over request alert modality.

With respect to the visual modality-based studies, Walch et al. [10] showed it was effective to vary the alert times using visual modality and give a take-over request 6 s

prior to an incident. Eriksson et al. [11] found in their study that an arrow form was the most effective for successful take-over to human drivers.

Among the auditory modality studies, the study by Naujoks et al. [12] showed that, in the event of take-over request in autonomous vehicles, speech was preferred over simple tone. Hester et al. [13] analyzed drivers' trust on warnings in mutually different types in the event of a vehicle failure in autonomous vehicles and found that more drivers successfully avoided collision when they had received speech warning describing emergency.

Among the tactile modality studies, the study by Petermeijuer et al. [14] looked into drivers' reaction time, availability, and pleasantness in relation to tactile modality in the event of a take-over request in autonomous vehicles. They found that, rather than alerting solely based on tactile modality, alerting via both tactile modality and auditory modality was more effective. Telpaz et al. [15] also studies whether a tactile modality alert through the seat improved driver's awareness to achieve effective take-over. They found that the tactile modality helped drivers react to take-over request more swiftly.

### 3.3 Alert Modality of Mass-Produced Autonomous Vehicle

Prior to establishing a take-over request alert methodology for conditional driving autonomous vehicle, the alert methods of partial autonomous vehicles (e.g., level 2 in [2]) produced by 8 manufacturers were investigated. As of now, there are only mass-produced vehicles at the SAE L2 and one manufacturer, Tesla Motors, provides a take-over request alert function.

In the event of failure to keep the distance between cars or lane departure, the visual modality-based alert devices in the presently mass-produced partial autonomous vehicles provide warnings through an icon or light on their instrument panel or HUD (Head-up Displayer). Next, the auditory modality-based alert interface sounded warning at a set frequency band in a set period if a driver does not operate the steering wheel for a long while or driving speed exceeds 37 miles per hour. A8, produced by Audi, alerts driver if it exceeds 37 miles per hour. Lastly, the tactile modality-based alert interface warned by sending vibration at certain intervals through the steering wheel or seat motor in the event of lane departure or potential collision. Leaf, produced by Nissan, gives an alert if it exceeds 37 miles per hour.

### 3.4 Establishment of Uni-modal Alert Design

A set of uni-modal alert design was established, which can be considered as the top priority in take-over request studies, based on the investigation on the guidelines on alert methods of existing non-autonomous vehicles, preceding studies on autonomous vehicles' alert methods and alert methods of autonomous vehicles mass-produced so far.

The visual-based alert method can provide a warning sign containing the phrase, "Take Over Immediately", on the head-up display in 3 Hz period until take-over is complete, as its first method described in Method 1 in Fig. 1. Subsequently, as shown in Method 2, it provides a warning sign containing the phrase, "Take Over Immediately" on the room mirror in 3 Hz period until the take-over is complete. Lastly, as in

Method 3, it can provide a warning icon to a bar-formed model with red-colored background and a warning sign including the phrase, “Take Over” on the instrument panel or head-up display in 3 Hz period until the take-over is completed.



**Fig. 1.** Example of visual-based alert method.

The auditory-based alert method borrows the warning sound of presently mass-produced partial autonomous vehicles and repeats 0.2-second-long warning sound two times at 0.06-s intervals at 533 Hz in the event of any take-over request incident; or provides 0.09-second-long warning sound three times at 0.03 intervals at 1498 Hz. It also employs these two warning sounds and the signal from previous study to mix 880 Hz and 1760 Hz single tones and sound 0.05-second-long warning three times at 0.03-s intervals.

In the tactile-based method, motor is horizontally arranged in the widest possible range around the back and gluteus of human drivers' seat and vibration alerts can be gradually increased from the lowest level of 60 Hz up to 250 Hz. Until the take-over is completed, 0.2-second-long pulse can be provided through the back and gluteus of human drivers' seat at 0.05-s intervals. Also, 0.2-second-long pulse at the lowest level of 200 Hz up to 250 Hz can be given through the gluteus of human drivers' seat at 0.05-s intervals.

## 4 Experimental Environment Setup

A driving simulator was constructed, which can materialize the take-over request scenario in virtual reality, provide take-over request alert to drivers, and measure their reactions.

Huynai's medium-sized sedan, LF SONATA, was utilized as a cabin. To implement the virtual driving environment and autonomous vehicle functions, the SCANeR 1.7 program was employed. To realize take-over request alert modality, visual, auditory and tactile alert interfaces were equipped. The visual modality alert interface was made with center fascia, instrument panel, rear-view mirror, and side mirrors LED monitor. The auditory modality alert interface was established by utilizing 5.1Ch speaker; and the tactile device, the vibration of seat, steering wheel, and accelerator pedal.

In order to measure drivers' status, NVR Camera was installed, which can provide video monitoring of driver behaviors. Drivers' sight patterns are identified through

Face Lab's eye tracker. In addition, BioPac's bio-signal interface was employed to gain drivers' bio-signals such as electrocardiogram, and GSR (Galvanic Skin Response).

## 5 Conclusion

In this study, a previous study was performed on alert modalities for take-over request, which can secure take-over safety in conditional autonomous vehicles (e.g., level 3 in [2]) expected to be commercialized.

The established take-over request scenario and multi-modality combination proposals will be materialized with the virtual environment simulator to build an experiment device capable of assessing drivers' reaction and gain vehicle and driver data. We plan to implement a take-over request alert experiment, draw quantitative/qualitative data, analyze statistically significant difference in driver and vehicle data according to take-over request multi-modal alert method, and establish a take-over request multi-modal alert method.

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